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Cycle embedding in star graphs with edge faults

Tseng-Kuei Li

*Department of Computer Science and Information Engineering, Ching Yun University,
JungLi, Taiwan 320, ROC*

Abstract

The star graph is one of the famous interconnection networks. Edge fault tolerance is an important issue for networks, and the cycle embedding problem is widely discussed in many researches. In this paper, we show that the cycles of even lengths from 6 to $n!$ can be embedded into the n -dimensional star graph when the number of edge fault in the graph does not exceed $n - 3$. Since the graph is bipartite and $(n - 1)$ regular, our result is optimal.

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1. Introduction

Network topology is a crucial factor for a network since it determines the performance of the network. For convenience of discussing their properties, networks are usually represented by graphs. In this paper, a network topology is represented by a simple undirected graph, which is loopless and without multiple edges. For the graph definition and notation we follow [4]. $G = (V, E)$

E-mail address: tealee@ms8.hinet.net

is a graph if V is a finite set and E is a subset of $\{(a,b)|a \neq b \in V\}$, where (a,b) denotes an unordered pair. We call V the *vertex set* and E the *edge set*. We say that vertices a and b are adjacent if and only if $(a,b) \in E$. The graph G is bipartite if and only if $V = V_0 \cup V_1$, $V_0 \cap V_1 = \emptyset$, and $E \subseteq \{(u,v)|u \in V_0, v \in V_1\}$. A path is a sequence of adjacent vertices, denoted by $\langle v_0, v_1, \dots, v_k \rangle$, in which $v_0, v_1, \dots, v_k$ are distinct except that possibly $v_0 = v_k$. The length of the path is k . We say that the path is a cycle if $v_0 = v_k$. A path (or a cycle) is hamiltonian with respect to a graph G if it crosses all vertices of G .

To find a cycle of given length in a graph is a cycle embedding problem. Some of studies focused on the longest cycle embedding problem. Other studies focused on the problem of finding cycles of all lengths in a graph, i.e., finding cycles of lengths from 3 to the number of vertices in the graph. We call the problem *the pancyclic problem*. In this paper, we focus on such a problem on the star graphs proposed by Akers et al. [1]. Since the star graphs are bipartite graphs, they contain no odd length cycles. Moreover, the minimum length of cycles of the star graphs is 6. Jwo et al. [10] showed that an n -dimensional star graph contains cycles of even length from 6 to $n!$. Thus, we say that the star graphs are weak-bipancyclic.

The star graphs [1] are Cayley graphs. They have many nice properties such as recursiveness, vertex and edge symmetry, maximal fault tolerance, sublogarithmic degree and diameter [1,5]. These properties are important for designing interconnection topologies for parallel and distributed systems. Star graphs are able to embed cycles, grids [10], trees [2], and hypercubes [14]. Many efficient communication algorithms for shortest-path routing [15], multiple-path routing [5], broadcasting [13], gossiping [3], and scattering [7] were proposed. And many efficient algorithms designed for sorting and merging [12], selection [15], Fourier transform [6], and computational geometry [16] have been proposed. As a result, star graphs are recognized as an attractive alternative to the hypercubes.

However, there is no result about such a property for faulty star graphs. Since components in a network would sometimes fail, to study graphs with faults is more practical. In this paper, we discuss the even-length-cycle embedding problem on the edge fault star graphs. We show that an n -dimensional star graph contains cycles of even lengths from 6 to $n!$ when the number of edge faults in the graph does not exceed $n - 3$. The result is optimal since the graph is $(n - 1)$ regular and $(n - 2)$ faults will cause the graph containing no hamiltonian cycle.

In the next section, we first introduce the definition of star graphs. In Section 3, we show the main result. And finally, we present a conclusion.

2. Definition and basic properties

The following is the definition of the star graphs. For convenience, we use $v_1 v_2 \dots v_n$ to represent vertex v in S_n in this paper.

Definition 1. The n -dimensional star graph, denoted by S_n , is the graph (V, E) , where $V = \{v | v \text{ is a permutation of } 1, 2, \dots, n\}$ and $E = \{(u, v) | v = u_i u_2 \dots u_{i-1} u_1 u_{i+1} \dots u_n\}$.

By this definition, S_n contains $n!$ vertices and each vertex is of degree $(n - 1)$. For example, 1234 is a vertex in S_4 and connects to 2134, 3214, and 4231. We use $N(u)$ to denote the neighborhood of u , e.g., $N(1234) = \{2134, 3214, 4231\}$. S_1 , S_2 , and S_3 are a vertex, an edge, and a cycle of length 6, respectively. We show S_4 in Fig. 1. It is clear that there are four vertex-disjoint S_3 s embedded in S_4 . The following lemma states this property.

Lemma 1. There are n vertex-disjoint S_{n-1} 's in S_n for $n \geq 2$.

Proof. Let i be an integer with $2 \leq i \leq n$. Let $H^{i,j} = (V^{i,j}, E^{i,j})$, where $V^{i,j} = \{u \in V(S_n) | u_i = j\}$ and $E^{i,j} = \{(u, v) \in E(S_n) | u, v \in V^{i,j}\}$ for $1 \leq j \leq n$. Clearly, $\{V^{i,1}, V^{i,2}, \dots, V^{i,n}\}$ is a partition of $V(S_n)$. Let $\pi(u) = u_1 u_2 \dots u_{i-1} u_{i+1} \dots u_n$. Then π is an isomorphism from $V^{i,j}$ to $V(S_{n-1})$. So $H^{i,j}$ is isomorphic to S_{n-1} . Thus, the lemma follows. $\square$

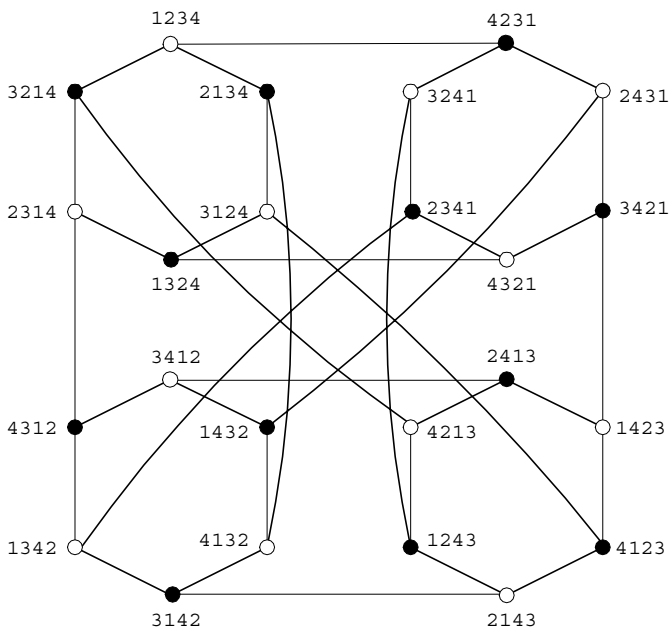


Fig. 1. Four-dimensional star graph.

We use $S_{n-1}^{i,j}$ to denote the subgraph $H^{i,j}$ in the above proof. Specifically, we use S_{n-1}^j as the abbreviation of $S_{n-1}^{n,j}$ and call it the j th $(n-1)$ -dimensional subgraph of S_n .

In the following, we introduce the property *k-edge-fault hamiltonian laceable*, which plays an important role in our main proof. Let $G = (V_0 \cup V_1, E)$ be a bipartite graph, where $V_0 \cap V_1 = \emptyset$ and $E \subseteq \{(u, v) | u \in V_0, v \in V_1\}$. Then G is *hamiltonian laceable* [8] if and only if, for any two vertices $x \in V_0$ and $y \in V_1$, there is a hamiltonian path of G between x and y . The graph G is *k-edge-fault hamiltonian laceable* if and only if G is still hamiltonian laceable when G contains at most k edge faults. The following theorem was proved by Hsieh et al. [9].

Theorem 1. S_n is $(n-3)$ -edge-fault hamiltonian laceable for $n \geq 4$.

By this theorem, we have the following result:

Lemma 2. Let $u, v \in V(S_n)$ with u and v in different partite sets. Then there are two paths of lengths $n! - 3$ and $n! - 1$, respectively, between u and v .

Proof. Since S_n is hamiltonian laceable, there is a path of length $n! - 1$ between u and v . Then consider the path of length $n! - 3$. Assume that $x \in N(v)$ with $x \neq u$ and $y \in N(x)$ with $y \neq v$ (see Fig. 2). Let $F = \{(x, z) | z \in N(x) - \{v, y\}\}$. Thus, $|F| = n - 3$, u and y are in different partite sets, and x only connects to v and y in $S_n - F$. Since S_n is $(n-3)$ -edge-fault tolerant hamiltonian laceable, there is a hamiltonian path between u and y in $S_n - F$. Clearly, the path should be of the form $\langle u, \dots, v, x, y \rangle$. Then we have a path of length $n! - 3$ in S_n between u and v . $\square$

Note that this result is only applied to the fault-free star graphs. We use this result in our main proof.

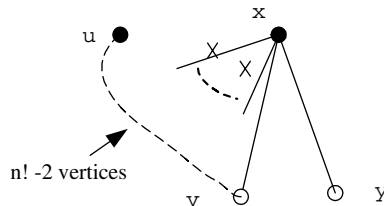


Fig. 2. Path of length $n! - 3$ between u and v .

3. Main result

In this section, we propose our main result, which is proven by induction. For the basis step, we enumerate all required cycles in the following lemma.

Lemma 3. *There is a cycle of each even length from 6 to $n!$ in $S_n - F$, where F is a set of edge faults with $|F| \leq n - 3$ for $n = 3$ and 4.*

Proof. For $n = 3$, S_3 is C_6 . Since $|F| = 0$, the lemma follows. For $n = 4$, $|F| = 1$. Since S_4 is edge-symmetric, we may assume that $F = \{(1243, 3241)\}$. Then we construct cycles of lengths 6, 8, 10, $\dots$, 24 in the following:

- C_6 : $\langle 1234, 4231, 2431, 1432, 4132, 2134, 1234 \rangle$.
- C_8 : $\langle 1234, 4231, 2431, 1432, 3412, 2413, 4213, 3214, 1234 \rangle$.
- C_{10} : Replace $\langle 4231, 2431 \rangle$ in C_6 by $\langle 4231, 3241, 2341, 4321, 3421, 2431 \rangle$.
- C_{12} : Replace $\langle 4231, 2431 \rangle$ in C_8 by $\langle 4231, 3241, 2341, 4321, 3421, 2431 \rangle$.
- C_{14} : Replace $\langle 1432, 4132 \rangle$ in C_{10} by $\langle 1432, 3412, 4312, 1342, 3142, 4132 \rangle$.
- C_{16} : Replace $\langle 1432, 3412 \rangle$ in C_{12} by $\langle 1432, 4132, 3142, 1342, 4312, 3412 \rangle$.
- C_{18} : Replace $\langle 2134, 1234 \rangle$ in C_{14} by $\langle 2134, 3124, 1324, 2314, 3214, 1234 \rangle$.
- C_{20} : Replace $\langle 2413, 4213 \rangle$ in C_{16} by $\langle 2413, 1423, 4123, 2143, 1243, 4213 \rangle$.
- C_{24} : Replace $\langle 3214, 1234 \rangle$ in C_{20} by $\langle 3214, 2314, 1324, 3124, 2134, 1234 \rangle$.
- C_{22} : Replace the subpath $\langle 4231, \dots, 3412 \rangle$ in C_{24} by $\langle 4231, 2431, 3421, 4321, 2341, 1342, 3142, 4132, 1432, 3412 \rangle$.

Hence, the lemma follows. $\square$

For the inductive step, we need to establish $(n - 2)$ disjoint paths crossing the given number of subgraphs to guarantee that there is still at least one of these paths crossing no faulty edge when the number of edge faults in S_n does not exceed $(n - 3)$. Moreover, the endpoints of established paths must be in the same subgraph. The following three lemmas discuss how to establish the disjoint paths. Then we use the paths to construct cycles of required lengths in the last lemma. For ease of description, we use u_i , u_i^j , and $(u^j)_i^k$ to denote the i th digits of vertices u , u^j , and $(u^j)^k$, respectively. We use P or $\langle v_0, P, v_k \rangle$ to denote the same path where the later points the two endpoints of path P . For two paths $P = \langle x, P, y \rangle$ and $Q = \langle u, Q, v \rangle$ with y and u being adjacent, we use $\langle x, P, y, u, Q, v \rangle$ to denote the path concatenating P with Q .

Lemma 4. *Let x and y be two distinct vertices in S_n for $n \geq 3$ with $x_1 = y_1$. Then $d(x, y) \geq 3$.*

Proof. By definition, $d(x, y) \neq 1$. Suppose that $d(x, y) = 2$. Then there is a vertex $z = z_1 z_2 \dots z_{n-1} z_n$ adjacent to x and y . So we have $x = z_i z_2 \dots$

$z_{i-1}z_1z_{i+1}\cdots z_n$ and $y = z_jz_2\cdots z_{j-1}z_1z_{j+1}\cdots z_n$ for some $2 \leq i, j \leq n$. Since $x \neq y$, $i \neq j$ and thus, $x_1 = z_i \neq z_j = y_1$ which contradicts the given condition. So $d(x, y) \geq 3$. $\square$

The following result directly follows from the above lemma.

Lemma 5. Let $x^1, x^2, \dots, x^{n-2}$ be distinct vertices in $V(S_{n-1}^i)$ for $n \geq 3$ with $x_1^1 = x_1^2 = \cdots = x_1^{n-2}$. And let u^j be a neighbor of x^j for $1 \leq j \leq n-2$. Then the $2(n-2)$ vertices are distinct.

Proof. By Lemma 4, $d(x^{i_1}, x^{i_2}) \geq 3$ for any $1 \leq i_1 \neq i_2 \leq n-2$. Suppose that u^r is x^t for some $1 \leq r \neq t \leq n-2$. Then $d(x^r, x^t) = 1$ which contradicts to Lemma 4. Suppose that u^r is u^t for some $1 \leq r \neq t \leq n-2$. Then $d(x^r, x^t) = 2$. We also get a contradiction. Thus, the lemma follows. $\square$

By this result, we may construct $(n-2)$ disjoint paths to link the assigned number of $(n-1)$ -dimensional subgraphs in S_n with all endpoints of the paths in the same subgraphs. First, we propose an algorithm to construct the $(n-2)$ paths and then show that the constructed paths are disjoint.

Algorithm 1

Input: m, n , where $n \geq 4$ and $3 \leq m \leq n$.

Output: $(n-2)$ disjoint paths.

1. Select arbitrary $(n-2)$ vertices, say $a^1, a^2, \dots, a^{n-2}$, from $V(S_{n-1}^m)$ with the first digits equal to 1// Note that the last bits of $a^1, a^2, \dots, a^{n-2}$ are m .
2. For $i \leftarrow 1$ to $(n-2)$
3. $P_i \leftarrow \langle a^i \rangle$
4. For $j \leftarrow 2$ to m
5. $v \leftarrow \text{lastVertex}(P_i)$ // Suppose that $v = v_1v_2\cdots v_{n-1}v_n$.
 // $\text{lastVertex}(P_i)$ returns the last vertex in P_i .
6. Append $v_nv_2\cdots v_{n-1}v_1$ to P_i
7. Append $v_lv_2\cdots v_{l-1}v_nv_{l+1}\cdots v_{n-1}v_1$ to P_i , where $v_l = j$
8. Append $v_nv_2\cdots v_{n-1}v_1$ to P_i , where v is the last vertex in P_i
9. Output $P_1, P_2, \dots, P_{n-2}$.

The following example illustrates the algorithm.

Example 1. Let $n = 6$ and $m = 3$. Suppose that we select $a_1 = 124563$, $a_2 = 125463$, $a_3 = 142563$, $a_4 = 145263$ in line 1. Then in lines 2–8, we obtain:

$\langle 124563, 324561, 234561, 134562, 314562, 214563 \rangle,$
 $\langle 125463, 325461, 235461, 135462, 315462, 215463 \rangle,$
 $\langle 142563, 342561, 243561, 143562, 341562, 241563 \rangle,$
 $\langle 145263, 345261, 245361, 145362, 345162, 245163 \rangle. \quad \square$

Since there are $(n-2)!$ vertices in S_{n-1}^m with the first digits equal to 1, it is not a hard work to find enough vertices in line 1. In line 3, each $P_i = \langle a_i \rangle$ is initialized. Consider lines 4–7. For each turn, we append two vertices into the tail of P_i , and the two vertices are in S_{n-1}^j for $2 \leq j \leq m$. After the last turn, the first digit of the last vertex in P_i is m . So, in line 8, the appended vertex is in S_{n-1}^m . Let $(v^i)^t$ be the t th vertex in P_i and $(v^i)_1^t$ and $(v^i)_n^t$ be the first digit and the last digit of $(v^i)^t$, respectively. Then we have $(v^i)_1^t = (v^i)_1^t$ and $(v^i)_n^t = (v^i)_n^t$ for all $1 \leq t \leq 2m$ and $1 \leq i, j \leq n-2$.

By Lemma 5, we can check that the $(n-2)$ paths are disjoint. We can also check that each path contains $2m$ vertices with two vertices in each subgraph, so each path crosses m subgraphs. Thus, we have the following result.

Lemma 6. *There are $(n-2)$ disjoint paths of length $(2m-1)$ crossing $S_{n-1}^1, S_{n-1}^2, \dots, S_{n-1}^m$ such that the endpoints of these paths are in S_{n-1}^m for $n \geq 3$ and $3 \leq m \leq n$.*

Clearly, in the above lemma, the endpoints of the paths can be in any $(n-1)$ -dimensional subgraph of S_n since S_n is vertex-symmetric. If the number of faults in S_n does not exceed $(n-3)$, the lemma may guarantee that there exists one of these paths being fault-free. The following lemma states the result. We use it in our main proof.

Corollary 1. *Let $F \subseteq E(S_n)$ with $|F| \leq n-3$ for $n \geq 3$. Then there is a path in $S_n - F$ of length $(2m-1)$ crossing $S_{n-1}^1, S_{n-1}^2, \dots, S_{n-1}^m$ for $3 \leq m \leq n$ such that the endpoints of the paths are in S_{n-1}^i for $1 \leq i \leq m$.*

Note that the two endpoints of the path are not adjacent, except for $m=3, 4$. Now we show the inductive step of our main result. For the convenience of description, we use *subgraphs* of S_n as $(n-1)$ -dimensional subgraphs of S_n .

Lemma 7. *There are cycles of all even lengths from 6 to $n!$ in $S_n - F$, where $F \subseteq E(S_n)$ with $|F| \leq n-3$ for $n \geq 5$.*

Proof. Assume that the statement is true for all $3 \leq k \leq n-1$. Since S_n is edge-symmetric, we may assume that there is a fault between two subgraphs. Then

each subgraph contains at most $(n-4)$ faults and is still hamiltonian laceable. Since $|F| \leq n-3$, we may assume that S_{n-1}^1 and S_{n-1}^n contain no fault.

By the hypothesis and each subgraph contains at most $(n-4)$ faults, we have cycles of all even lengths from 6 to $(n-1)!$ in any subgraph. Thus, we have cycles of even lengths from 6 to $(n-1)!$ in $S_n - F$.

Second, we construct cycles of even lengths from $(n-1)! + 2$ to $(n-1)! + 2(n-2)$. Suppose that $l = (n-1)! - 4 + 2m$ for some $3 \leq m \leq n$. By Corollary 1, there is a path P of length $2m-1$ crossing m subgraphs of $S_n - F$ with two endpoints in S_{n-1}^1 . Assume that the two endpoints are u and v . By Lemma 2, there is a path Q of length $(n-1)! - 3$ in S_{n-1}^1 between u and v . Then $\langle u, P, v, Q, u \rangle$ forms a cycle of length $(n-1)! - 4 + 2m$ in $S_n - F$.

Third, we construct cycles of even lengths from $(n-1)! + 2(n-1)$ to $n!$. Let l be an arbitrary even integer between $(n-1)! + 2(n-1)$ and $n!$. Then there exist two integers k and h such that $l = k((n-1)! - 2) + 2n + 2h$ for $1 \leq k \leq n$ and $0 \leq h < \frac{(n-1)!-2}{2}$. We construct cycles of length l in $S_n - F$ in the following:

By Corollary 1, there is a fault-free path $P = \langle u^1, v^2, u^2, v^3, \dots, v^n, u^n, v^1 \rangle$ in $S_n - F$ such that $u^i, v^i \in V(S_{n-1}^i)$. The number of vertices on P is $2n$. Then we add vertices into P by the following steps. After all steps, we obtain a cycle of length l :

Step 1: Add $(k-1)((n-1)! - 2)$ vertices.

If $k \geq 2$, replace each edge (v^i, u^i) by a hamiltonian path of $S_{n-1}^i - F$ for $2 \leq i \leq k$.

If $k = 1$, do nothing.

Step 2: Add $2h$ vertices if $h \neq 1$ or $2h+2$ vertices if $h = 1$.

If $h \geq 2$, replace the edge (v^n, u^n) by a path of length $2h+1$ in S_{n-1}^n .

If $h = 1$, replace the edge (v^n, u^n) by a path of length 5 in S_{n-1}^n .

If $h = 0$, do nothing.

Step 3: Add $((n-1)! - 2)$ vertices if $h \neq 1$ or $((n-1)! - 4)$ vertices if $h = 1$.

If $h \neq 1$, add a hamiltonian path $\langle u^1, Q', v^1 \rangle$ of S_{n-1}^1 into P .

If $h = 1$, add a path $\langle u^1, Q', v^1 \rangle$ of length $(n-1)! - 3$ into P .

Note that since there is no cycle of length 4 in the star graph, we cannot add $2h = 2$ vertices into path P in Step 2 when $h = 1$. Thus, we add 4 vertices into P in Step 2 and then add $(n-1)! - 4$ vertices in Step 3 when $h = 1$.

Now we check the executability of all steps. Step 1 is executable since all substars are still hamiltonian laceable, and since S_{n-1}^1 contains no fault, by Lemma 2, Step 3 is also executable. Consider Step 2. S_{n-1}^n is assumed to be fault-free. Since S_{n-1} is edge symmetric and, by hypothesis, has cycles of all even length from 6 to $(n-1)!$, there exists a cycle in S_{n-1}^n crossing (v^n, u^n) and of length $2h+2$ if $h \neq 1$. On the other hand, if $h = 1$, we can find a cycle in S_{n-1}^n of

length 6 crossing (v'', u'') . Then we can obtain the needed path by removing (v'', u'') from the cycle.

Hence, the lemma follows. $\square$

So we have the following result:

Theorem 2. *There are cycles of all even length from 6 to $n!$ in $S_n - F$, where F is a set of edge faults with $|F| \leq n - 3$ for $n \geq 3$.*

4. Conclusion

In this paper, we show that the n -dimensional star graph, S_n , contains cycles of all even length from 6 to $n!$ when there are some edges which have failed if the number of faulty edges does not exceed $(n - 3)$. Moreover, if there are $(n - 2)$ faulty edges incident to some vertex, the faulty S_n contains no cycle of length $n!$. Thus, our result is optimal.

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